DIFFUSION IN SOLIDS

- FICK'S LAWS
- KIRKENDALL EFFECT
- ATOMIC MECHANISMS

Part of
MATERIALS SCIENCE & ENGINEERING *A Learner's Guide*
AN INTRODUCTORY E-BOOK

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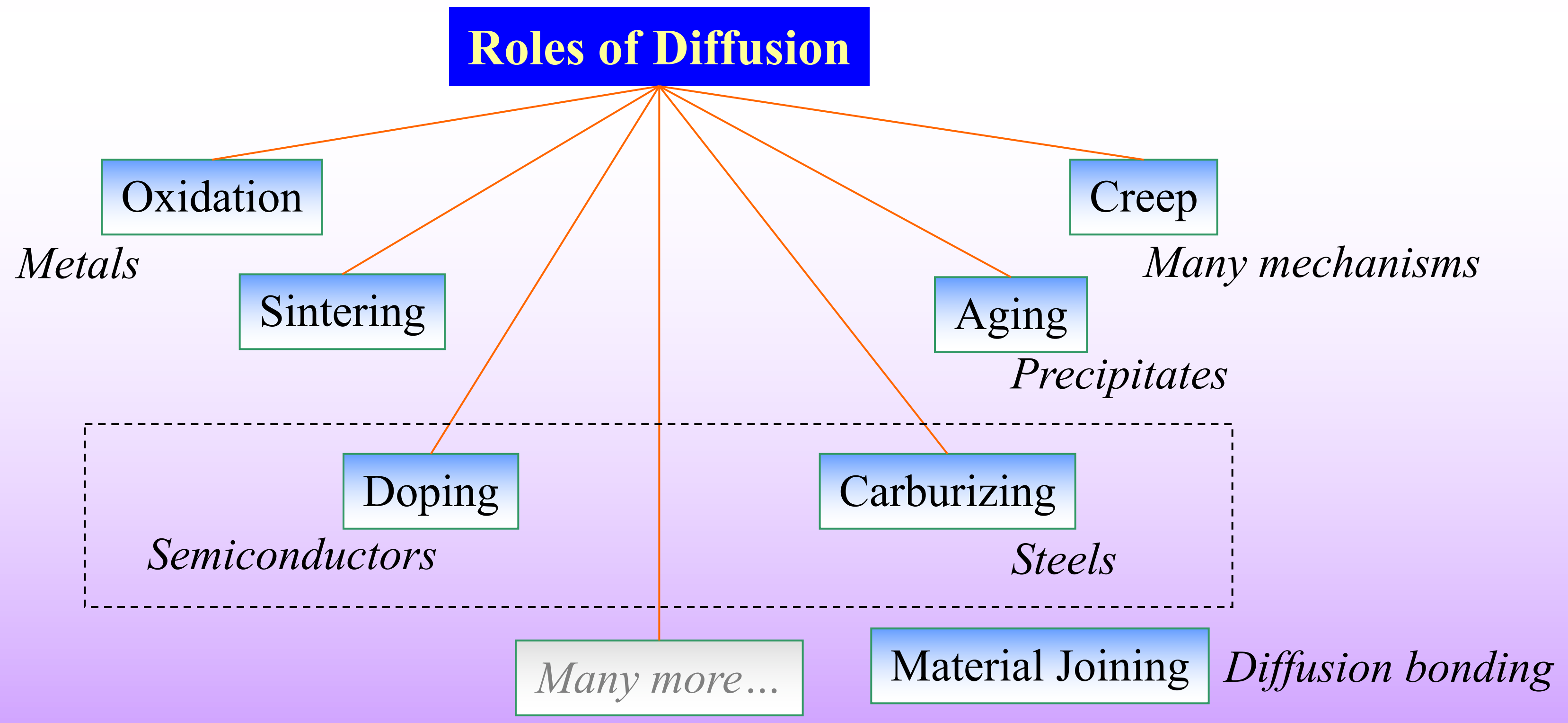
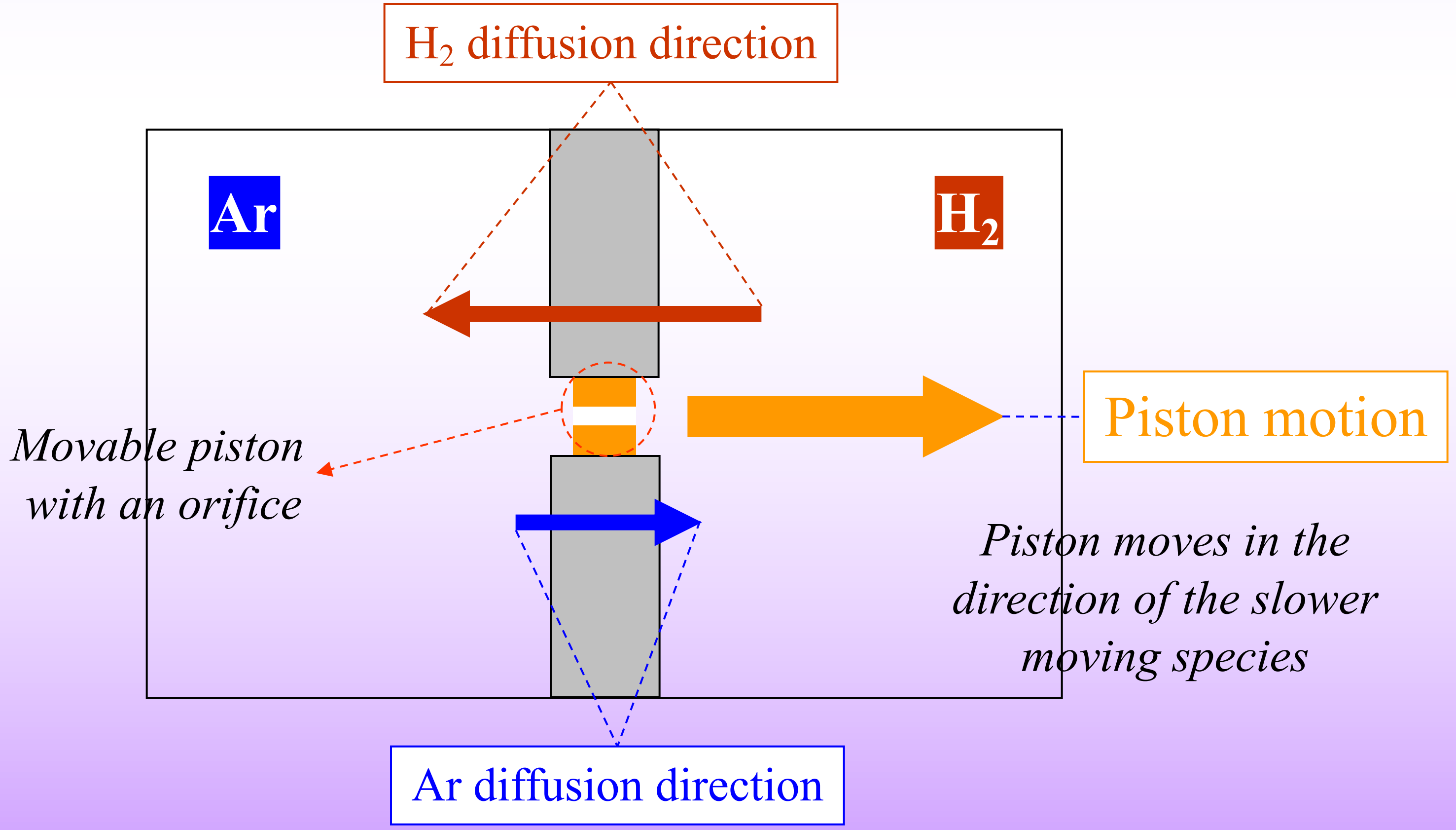
Diffusion in Solids
P.G. Shewmon
McGraw-Hill, New York (1963)

- Diffusion is relative flow of one material into another
  - *Mass flow process by which species change their position relative to their neighbours.*
- Diffusion of a species occurs from a region of high concentration to low concentration (usually). More accurately, diffusion occurs down the chemical potential (μ) gradient.
- To comprehend many materials related phenomenon (as in the figure below) one must understand Diffusion.
- The focus of the current chapter is solid state diffusion in crystalline materials.
- In the current context, diffusion should be differentiated with flow (of usually fluids and sometime solids).

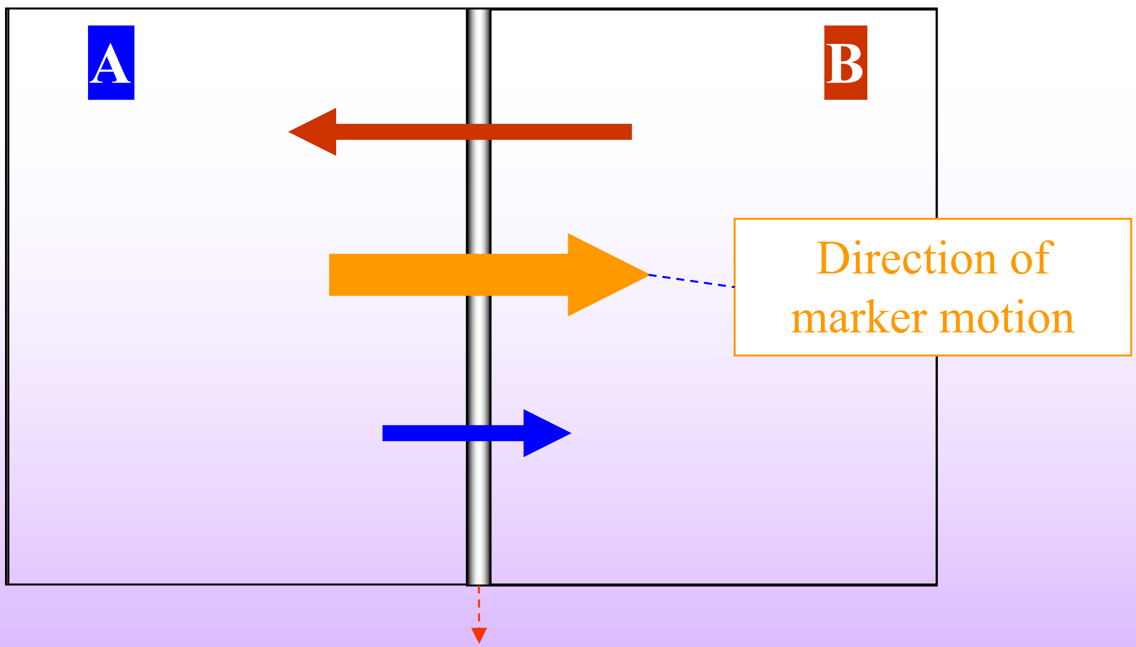


- When a perfume bottle is opened at one end of a room, its smell reaches the other end via the diffusion of the molecules of the perfume.
- If we consider an experimental setup as below (with Ar and H_2 on different sides of a chamber separated by a movable piston), H_2 will diffuse faster towards the left (as compared to Ar). As obvious, this will lead to the motion of movable piston in the direction of the slower moving species.
- This experiment can be used to understand the Kirkendall effect.



Kirkendall effect

- Let us consider two materials A and B welded together with Inert marker and given a diffusion anneal (i.e. heated for diffusion to take place).
- Usually the lower melting component diffuses faster (say B). This will lead to the shift in the marker position to the right.
- This is called the Kirkendall effect.

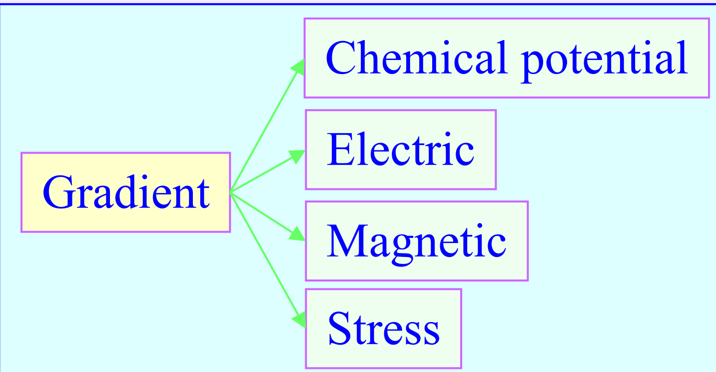


Inert Marker is basically a thin rod of a high melting material, which is insoluble in A & B

Diffusion

- Mass flow process by which species change their position relative to their neighbours.
- Diffusion is driven by *thermal energy* and a '*gradient' (usually in chemical potential).* Gradients in other physical quantities can also lead to diffusion (as in the figure below). In this chapter we will essentially restrict ourselves to concentration gradients.
- Usually, *concentration gradients imply chemical potential gradients*; but there are exceptions to this rule. Hence, sometimes diffusion occurs 'uphill' in concentration gradients, but downhill in chemical potential gradients.
- Thermal energy leads to thermal vibrations of atoms, leading to atomic jumps.
- In the absence of a gradient, atoms will still randomly 'jump about', without any net flow of matter.

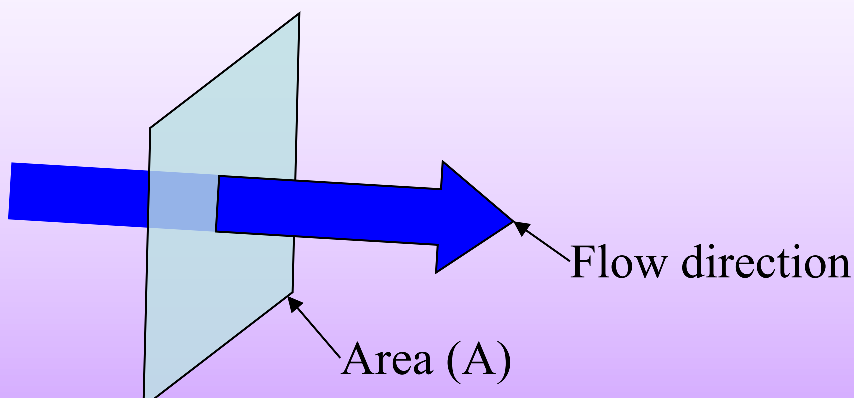
- First we will consider a *continuum picture* of diffusion and later consider the *atomic basis* for the same in crystalline solids. The continuum picture is applicable to heat transfer (i.e., is closely related to mathematical equations of heat transfer).



Important terms

- Concentration gradient. Concentration can be designated in many ways (e.g. moles per unit volume). Concentration gradient is the difference in concentration between two points (usually close by).
- We can use a restricted definition of flux (J) as flow per unit area per unit time:
  → mass flow / area / time → [Atoms / m^2 / s].
- Steady state. The properties at a single point in the system does not change with time. These properties in the case of fluid flow are pressure, temperature, velocity and mass flow rate.
  ▪ In the context of diffusion, steady state usually implies that, concentration of a given species at a given point in space, does not change with time.

- In diffusion problems, we would typically like to address one of the following problems.
  (i) What is the composition profile after a contain time (i.e. determine c(x,t))?



$$J = \frac{1}{A}\frac{dn}{dt}$$

$$J \to \left[\frac{mass}{area \cdot time}\right] = \left[\frac{atoms}{m^2 \cdot s}\right]$$

Fick's* I law

As we shall see the 'law' is actually an equation

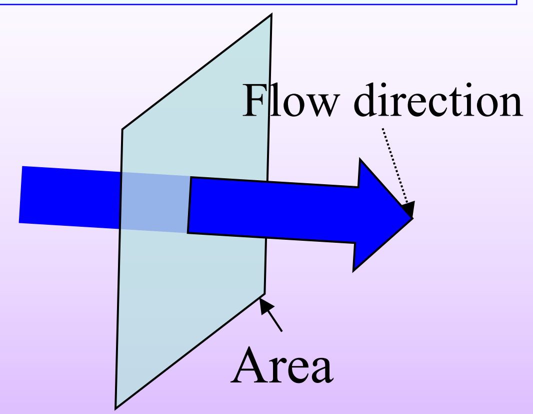
- Assume that only species 'S' is moving across an area 'A'. Concentration gradient for species 'S' exists across the plane.
- The concentration gradient *(dc/dx)* drives the flux *(J)* of atoms.
- Flux (J) is assumed to be proportional to concentration gradient.
- The constant of proportionality is the Diffusivity or Diffusion Coefficient (D).
  - 'D' is assumed to be independent of the concentration gradient.
  - Diffusivity is a material property. It is a function of the composition of the material and the temperature.
  - In crystals with cubic symmetry the diffusivity is isotropic (i.e. does not depend on direction).
- Even if steady state conditions do not exist (concentration at a point is changing with time, there is accumulation/depletion of matter), Fick's I-equation is *still valid* (but not easy to use).

The negative sign implies that diffusion occurs down the concentration gradient

A material property

$$J \propto \frac{dc}{dx} \rightarrow J = -D\frac{dc}{dx} \rightarrow J = \frac{1}{A}\frac{dn}{dt} = -D\frac{dc}{dx} \rightarrow \frac{dn}{dt} = -DA\frac{dc}{dx}$$

Fick's first law (equation)



* *Adolf Fick in 1855*

Let us emphasize the terms in the equation

$$\frac{dn}{dt} = -DA\frac{dc}{dx}$$

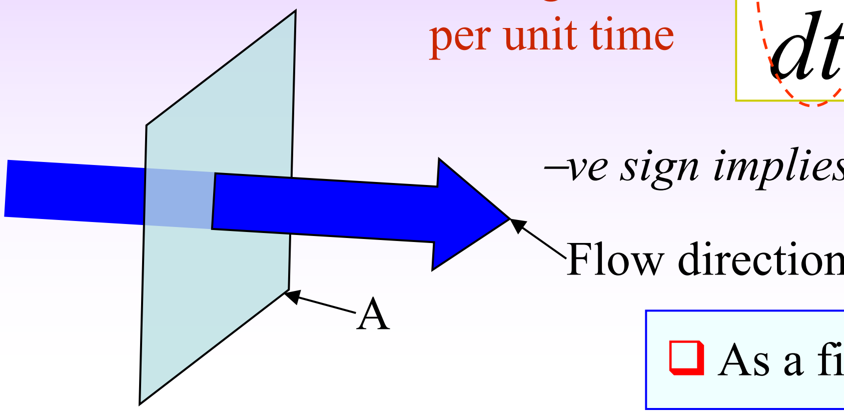
No. of atoms crossing area A per unit time → $\frac{dn}{dt}$

Diffusion coefficient/Diffusivity → D

Cross-sectional area → A

Concentration gradient → $\frac{dc}{dx}$

–ve sign implies matter transport is down the concentration gradient



- As a first approximation assume D ≠ f(t)

Let us look at the units of Diffusivity

$$\left\{ J = -D\frac{dc}{dx} \right\} \rightarrow \left[\frac{number}{m^2\, s}\right] = [D]\left[\frac{number}{m^3}\cdot\frac{1}{m}\right]$$

$$\left[\frac{m^2}{s}\right] = [D]$$

Note the strange unit of D: [m^2/s]

Fick's II law

Another equation

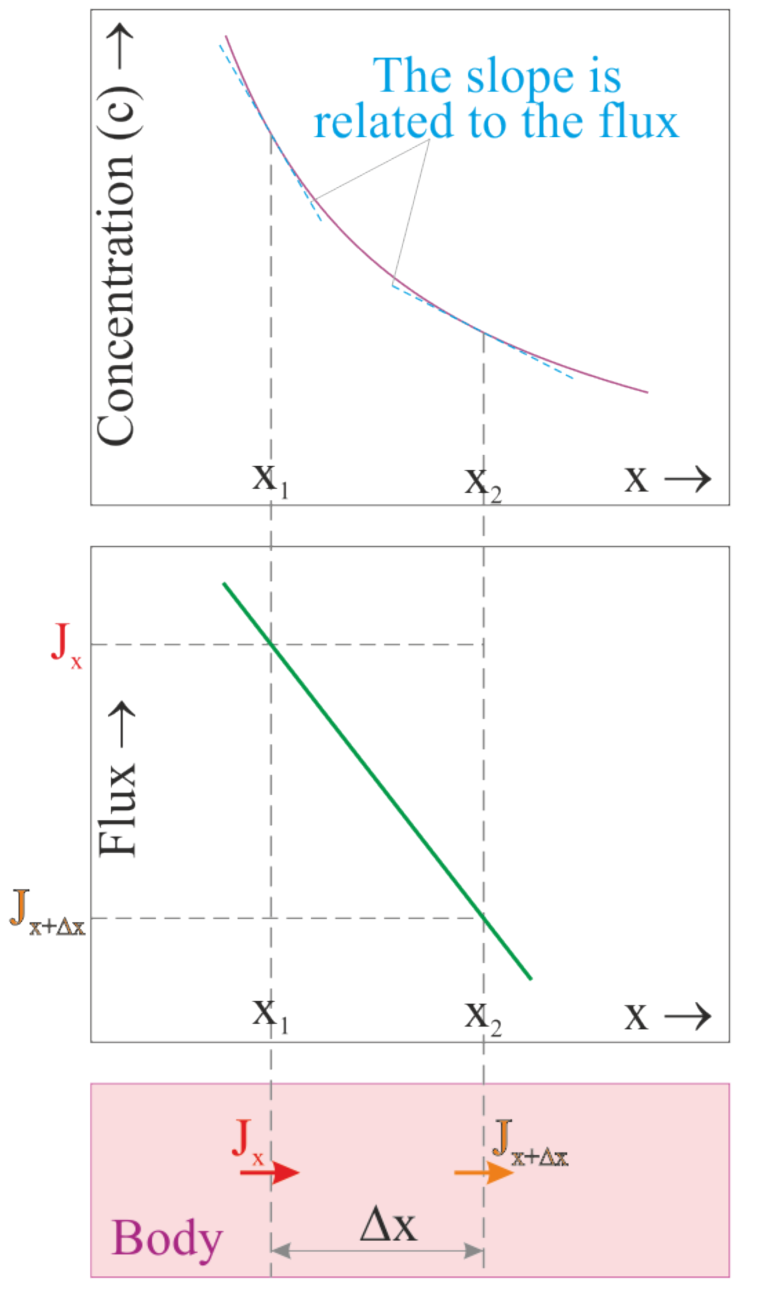
- The equation as below is often refered to as the Fick's II law (though clearly this is an equation and not a law).
- This equation is derived using Fick's I-equation and mass balance.
- The equation is a second order PDE requiring one initial condition and two boundary conditions to solve.

$$\boxed{\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right)}$$

Derivation of this equation will taken up next.

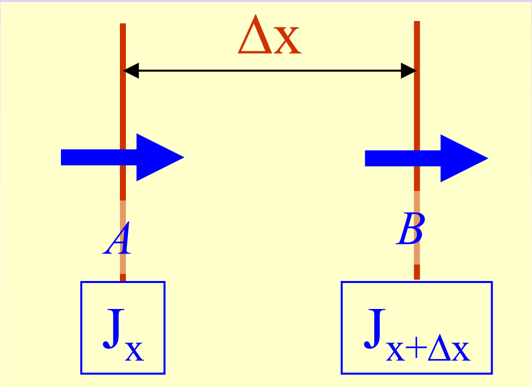
- If 'D' is not a function of the position, then it can be 'pulled out'.

$$\boxed{\left(\frac{\partial c}{\partial t}\right) = D\frac{\partial^2 c}{\partial x^2}}$$



- Let us consider a 1D diffusion problem.
- Let us consider a small element of width Δx in the body.
- Let the volume of the element be the control volume (V) = 1.1. $\Delta x = \Delta x$. *(Unit height and thickness).*
- Let the concentration profile of a species 'S' be as in the figure.
- The slope of the c-x curve is related to the flux via the Fick's I-equation.
- In the figure the flux is decreasing linearly.
- The flux entering the element is J_x and that leaving the element is $J_{x+\Delta x}$.
- Since the flux at x_1 is not equal to flux leaving that leaving at x_2 and since $J(x_1) > J(x_2)$, there is an accumulation of species 'S' in the region Δx.
- The increase in the matter (species 'S') in the control volume (V) = $(\partial c/\partial t).V = (\partial c/\partial t).\ \Delta x$.

- If J_x is the flux arriving at plane A and $J_{x+\Delta x}$ is the flux leaving plane B. Then the Accumulation of matter is given by: $(J_x - J_{x+\Delta x})$.



$$Accumulation = J_x - J_{x+\Delta x}$$

$$Accumulation = J_x - \left\{ J_x + \frac{\partial J}{\partial x}\Delta x \right\}$$

$$\left(\frac{\partial c}{\partial t}\right)\Delta x = J_x - \left\{ J_x + \frac{\partial J}{\partial x}\Delta x \right\}$$

Calculation of units

$$\left[\left(\frac{Atoms}{m^3}\frac{1}{s}\right).m\right] = \left[\frac{Atoms}{m^2 s}\right] \equiv [J]$$

$$\left(\frac{\partial c}{\partial t}\right)\Delta x = -\frac{\partial J}{\partial x}\Delta x \rightarrow \left(\frac{\partial c}{\partial t}\right) = -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right)$$

Fick's first law

$$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right) \xrightarrow{D \neq f(x)} \left(\frac{\partial c}{\partial t}\right) = D\frac{\partial^2 c}{\partial x^2}$$

Arises from mass conservation (hence not valid for vacancies)

In 3D

$$\left(\frac{\partial c}{\partial t}\right) = -\nabla J$$

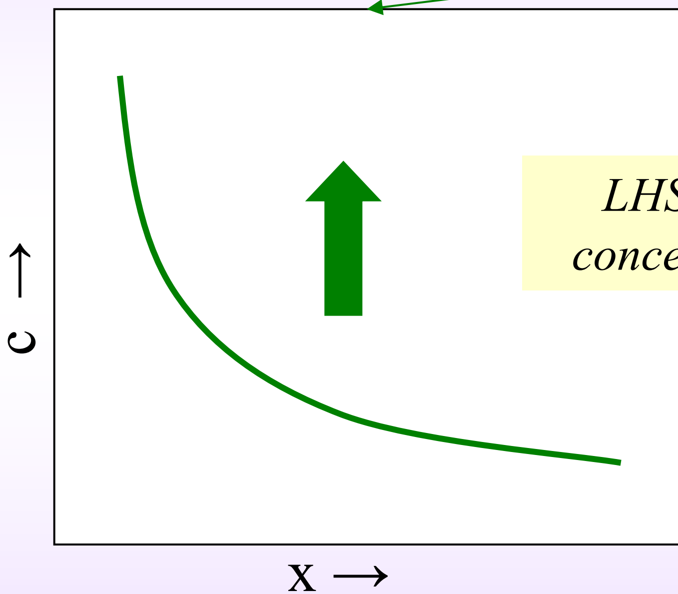
In 3D

$$\left(\frac{\partial c}{\partial t}\right) = D\nabla^2 c$$

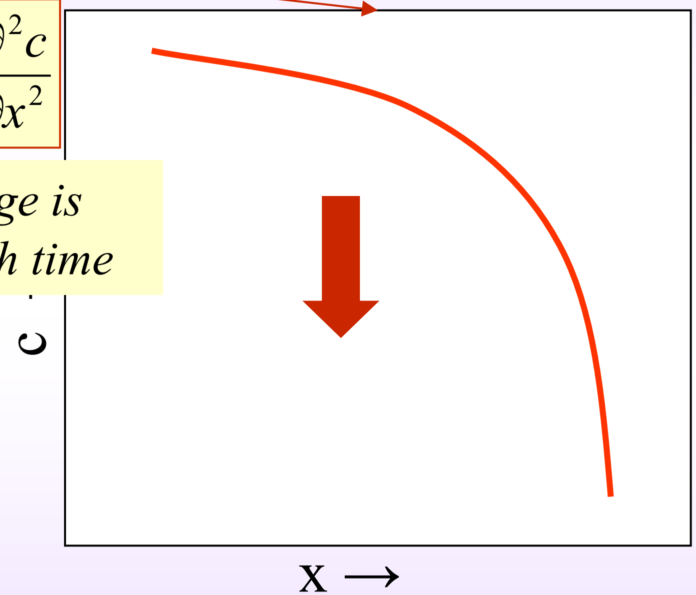
RHS is the curvature of the c vs x curve

$$\left(\frac{\partial c}{\partial t}\right) = D\frac{\partial^2 c}{\partial x^2}$$

LHS is the change is concentration with time



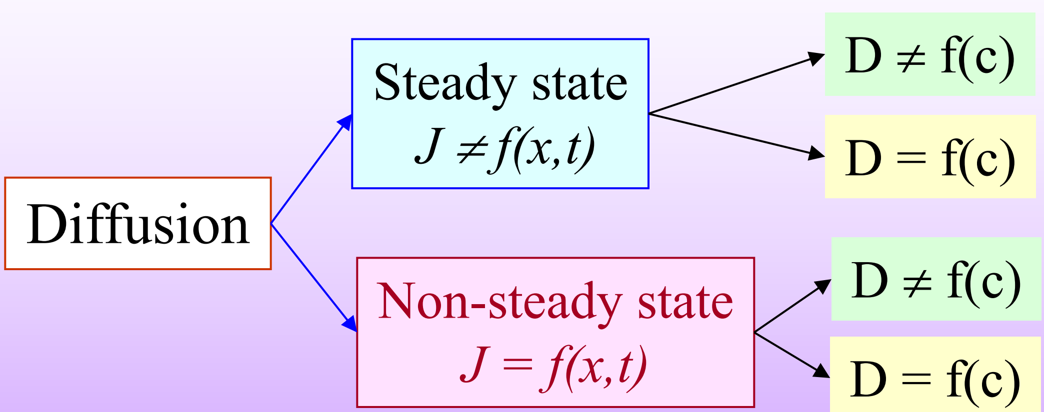
+ve curvature ⇒ c ↑ as t ↑



−ve curvature ⇒ c ↓ as t ↑

Steady and non-steady state diffusion

- Diffusion can occur under steady state or non-steady state (transient) conditions.
- Under steady state conditions, the flux is not a function of the position within the material or time. Under non-steady state conditions this is not true.
- This implies that under steady state the concentration profile does not change with time.
- In each of these circumstances, diffusivity (D) may or may not be a function of concentration (c). The term *concentration* can also be replaced with *composition*.



The general form of the Fick's II-equation is: $$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right)$$ In 3D $\rightarrow$ $$\left(\frac{\partial c}{\partial t}\right) = -\nabla J$$

- The equation is a second order differential equation involving time and one spatial dimension.
- This equation can be simplified for various circumstances and solved, as we will consider one by one. These include: (i) steady state conditions and (ii) non-steady state conditions.

Under steady state conditions

$$\left(\frac{dc}{dt}\right)_x = -\left(\frac{\partial J}{\partial x}\right)_t = 0$$

Substituting for flux from Fick's first law

$$-\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right) = 0$$

If D is constant $\rightarrow$

$$D\frac{\partial^2 c}{\partial x^2} = 0$$

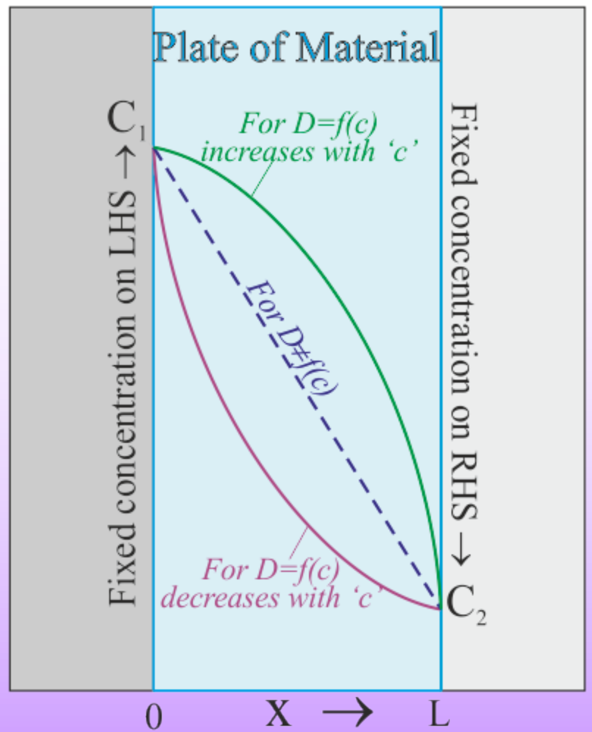
$\Rightarrow$ *Slope of c-x plot is constant under steady state conditions*

If D is NOT constant $\downarrow$

$$D\frac{\partial c}{\partial x} = \text{constant}$$

$\Rightarrow$ *If D increases with concentration then slope (of c-x plot) decreases with 'c'*

$\Rightarrow$ *If D decreases with 'c' then slope increases with 'c'*



Under non-steady state conditions

$$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right) \xrightarrow[\text{function of position}]{\text{If D is not a}} \left(\frac{\partial c}{\partial t}\right) = D\frac{\partial^2 c}{\partial x^2} \xrightarrow{\text{In 3D}} \left(\frac{\partial c}{\partial t}\right) = D\nabla^2 c$$

- The first simplification we make for the non-steady state conditions is that 'D' is not a function of the position.
- If the diffusion distance is short relative to dimensions of the initial inhomogeneity, we can use the error function (*erf*) solution with 2 arbitrary constants.
- The constants can be solved for from Boundary Condition(s) and Initial Condition(s). (we will take up examples to clarify this).

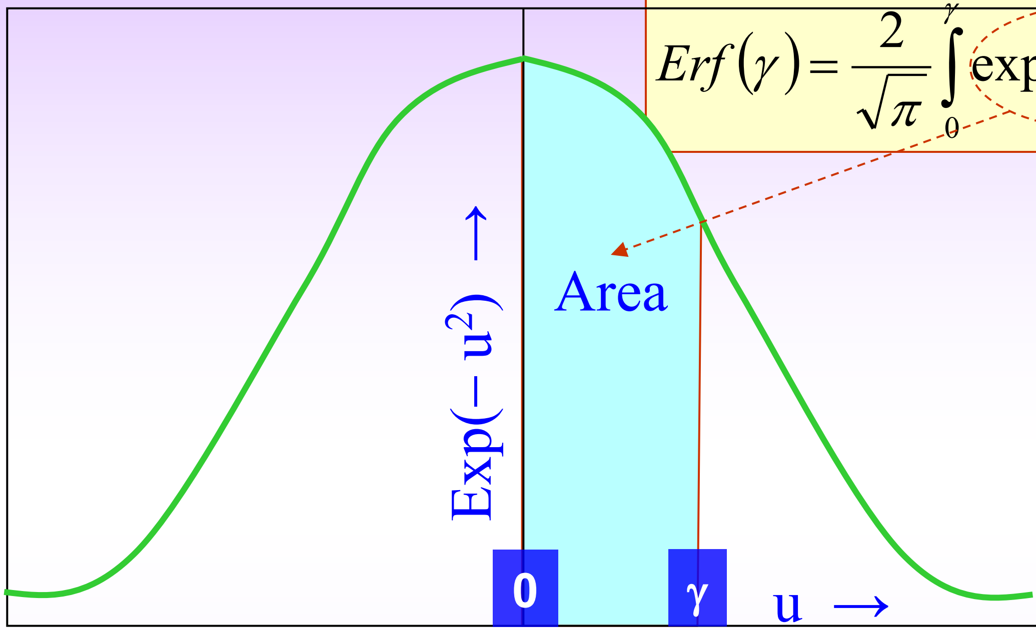
$$c(x,t) = A - B\,erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

- Under other conditions other solutions can be applied. For example, if a fixed amount of material is deposited on the surface of an infinite body and diffusion is allowed to take place, the concentration profile can be determined from the function below.

$$c(x,t) = \frac{M}{\sqrt{\pi D t}}\exp\left[\frac{-x^2}{4Dt}\right]$$

The error function (erf(γ)) is defined as below. The modulus of the function represents the area under the curve of the *exp(–u²)* function between ‘0’ and γ (with ‘some’ constant scaling factor). Some properties of the error function are also listed below.

$$Erf(\gamma) = \frac{2}{\sqrt{\pi}} \int_0^{\gamma} \exp\left(-u^2\right) du$$



Properties of the error function

- $Erf(\infty) = 1$
- $Erf(-\infty) = -1$
- $Erf(0) = 0$
- $Erf(-x) = -Erf(x)$

Also

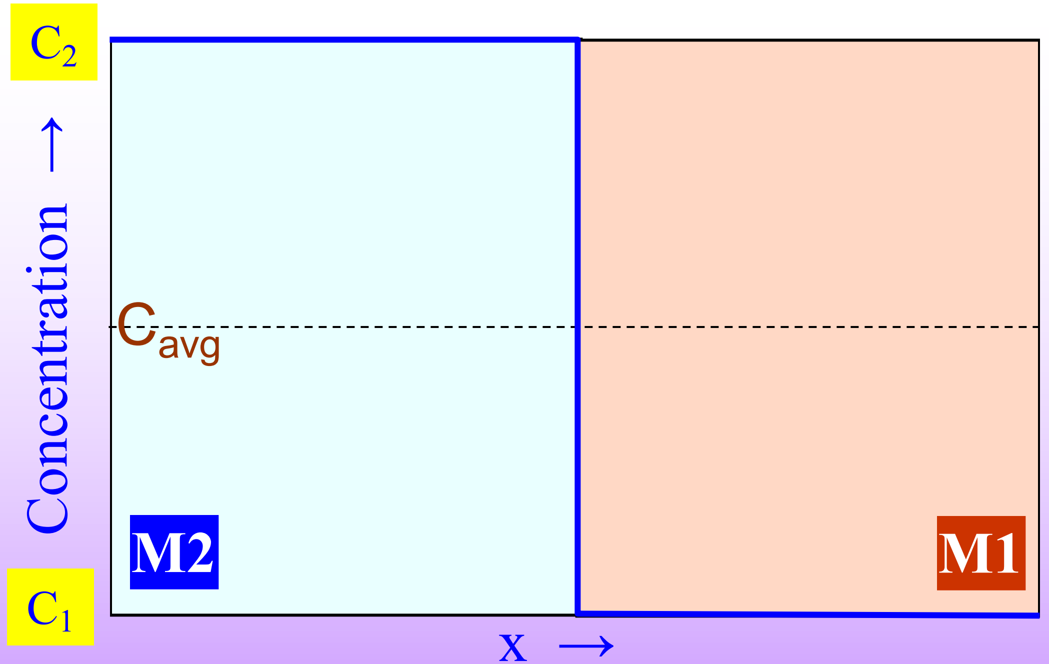
- For upto $x \sim 0.6 \rightarrow Erf(x) \sim x$
- $x \rightarrow 2,\ Erf(x) \rightarrow 1$

An example where the error function (erf) solution can be used

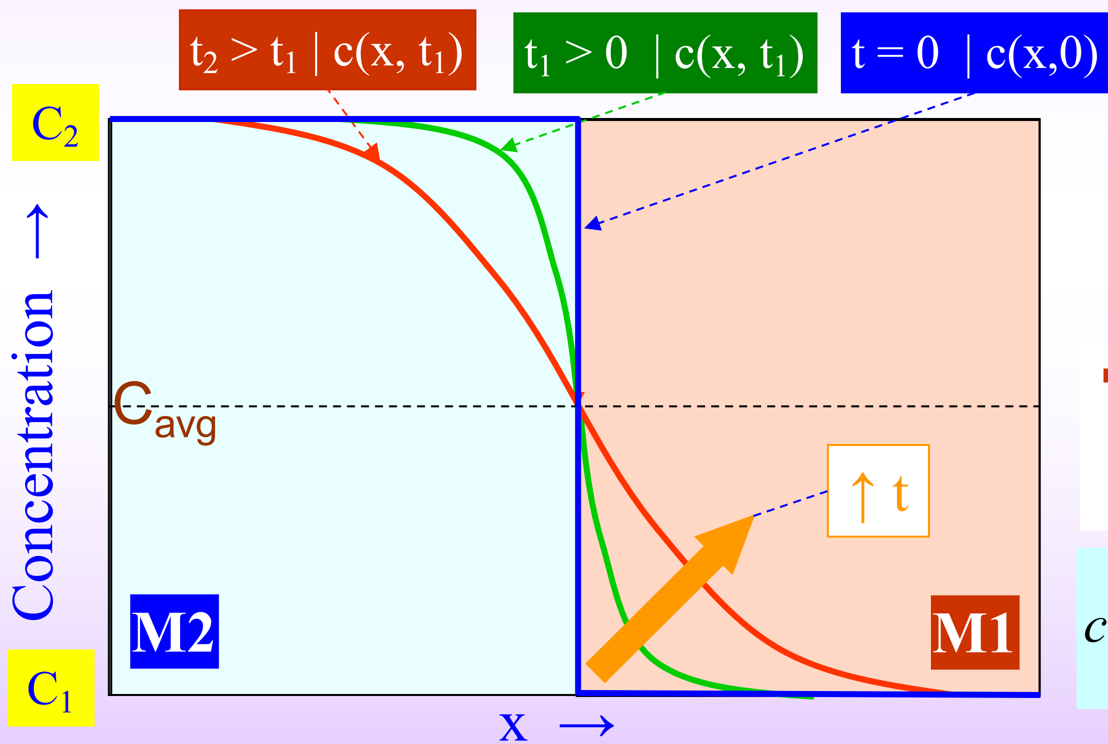
- Let two materials M1 & M2 be joined together and kept at a temperature (T_0), where diffusion is appreciable. Let C_1 be the concentration of a species in M1 and C_2 in M2.
- This is a 1D diffusion problem (i.e. the species diffuses along x-direction only).
- The initial concentration profile *(at t = 0, c(x,0)) of a species* is like a step function (blue line). If M1 and M2 are pure materials, then C_1 would be zero.
- We can define an average composition of the species as: $(C_1 + C_2)/2$.

The *initial conditions* (at t = 0) can be written as:

- $C(+x, 0) = C_1$
- $C(-x, 0) = C_2$



- With increasing time the species ‘S’ diffuses into M1 leading to a depletion of S in the region close to the interface on the M2-side and enrichment on the M1-side.
- This implies that we are dealing with non-steady state (transient) diffusion.
- From the initial conditions the arbitrary constants A & B can be determined and the concentration profile as a function of time (t) and position (x) can be determined.
- Such a profile for two specific times (t_1 and t_2) are shown below.



Flux → $f(x)|_t$, $f(t)|_x$ ➡ Non-steady state

- *If D = f(c)*
  $\Rightarrow c(+x,t) \neq c(-x,t)$
  i.e. asymmetry about y-axis

$$c(x,t) = A - B\,erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

- $C(+x, 0) = C_1$
- $C(-x, 0) = C_2$

→

- $A - B = C_1$
- $A + B = C_2$

→

- $A = (C_1 + C_2)/2$
- $B = (C_2 - C_1)/2$

$$c(x,t) = \left(\frac{C_1 + C_2}{2}\right) - \left(\frac{C_2 - C_1}{2}\right) erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

Temperature dependence of diffusivity

- Diffusion is an activated process and hence the Diffusivity depends exponentially on temperature (as in the Arrhenius type equation below).
- 'Q' is the activation energy for diffusion. 'Q' depends on the kind of atomic processes (i.e. mechanism) involved in diffusion (e.g. substitutional diffusion, interstitial diffusion, grain boundary diffusion, etc.).
- This dependence has important consequences with regard to material behaviour at elevated temperatures. Processes like precipitate coarsening, oxidation, creep etc. occur at very high rates at elevated temperatures.

Arrhenius type

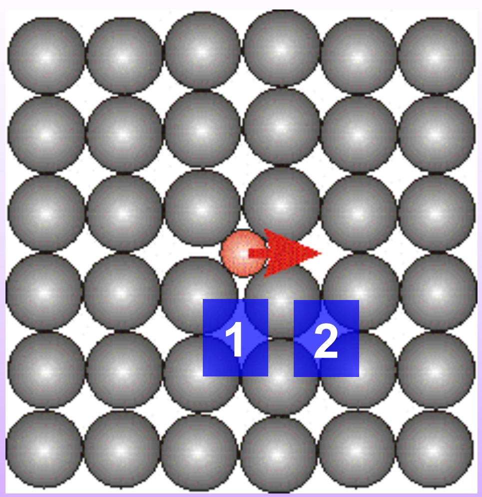
$$D = D_0\, e^{\left(-\frac{Q}{kT}\right)}$$

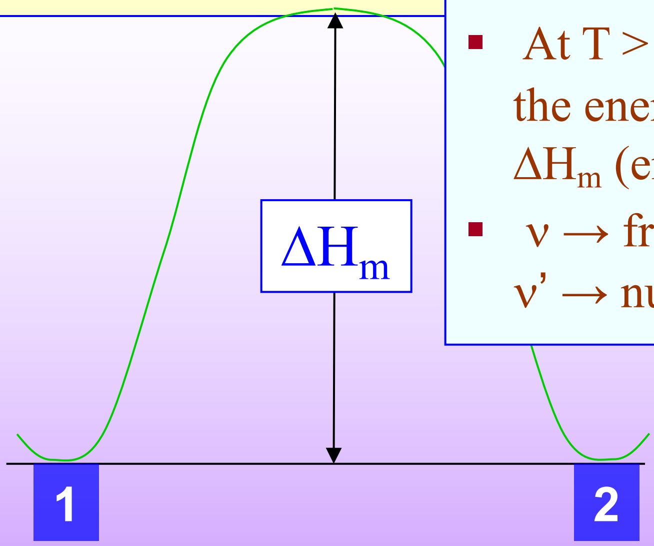
Atomic Models of Diffusion

- The diffusion of two important types of species needs to be distinguished:
  (i) species in a interstitial void (interstitial diffusion)
  (ii) species 'sitting' in a lattice site (substitutional diffusion).

1) Interstitial Diffusion

- Usually the solubility of interstitial atoms (e.g. carbon in steel) is small. This implies that most of the interstitial sites are vacant. Hence, if an interstitial species (like carbon) wants to jump, this is 'most likely' possible as the the neighbouring site will be vacant.
- Light interstitial atoms like hydrogen can diffuse very fast. For a correct description of diffusion of hydrogen anharmonic and quantum (under barrier) effects may be very important (especially at low temperatures).





- At T > 0 K vibration of the atoms provides the energy to overcome the energy barrier ΔH_m (enthalpy of motion).
- $\nu \rightarrow$ frequency of vibrations,
  $\nu' \rightarrow$ number of successful jumps / time.

$$\nu' = \nu \, e^{\left(-\frac{\Delta H_m}{kT}\right)}$$

2) Substitutional diffusion via Vacancy Mechanism

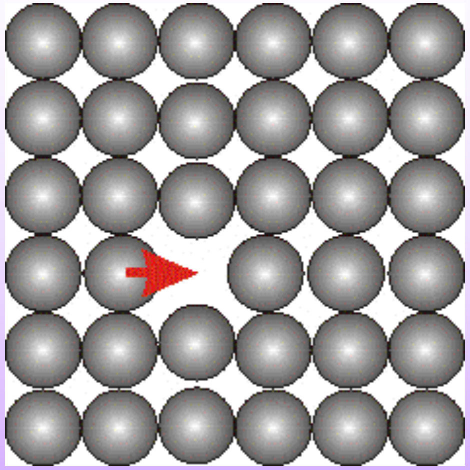
- For an atom in a lattice site (or a large atom associated with the motif), a jump to a neighbouring site can take place only if it is vacant. Hence, vacancy concentration plays an important role in the diffusion of species at lattice sites via the vacancy mechanism.
- Vacancy clusters and defect complexes can alter this simple picture of diffusion involving vacancies.

- *Probability for an atomic jump* $\propto$
  (probability that the site is vacant) $\times$ *(probability that the atom has sufficient energy)*

$$\nu' = \nu\, e^{\left(\frac{-\Delta H_f}{kT}\right)} e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

$$\nu' = \nu\, e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)}$$

- $\Delta H_m \rightarrow$ *enthalpy of motion of atom across the barrier.*
- $\nu' \rightarrow$ *frequency of successful jumps.*



Hence, ν' is of the form:

$$\nu' = \nu \, e^{\left(\frac{-\Delta(Enthalpy)}{kT}\right)}$$

If δ is the jump distance then the diffusivity can be written as:

$$D = \nu' \, \delta^2$$

$$D = \left\{ \nu \, e^{\left(\frac{-\Delta(Enthalpy)}{kT}\right)} \right\} \left\{ \delta^2 \right\}$$

- A comparison of the value of diffusivity for interstitial diffusion and substitutional diffusion is given below. The comparison is made for C in γ-Fe and Ni in γ-Fe (both at 1000°C).
- It is seen that $D_{interstitial}$ is orders of magnitude greater than $D_{substitutional}$.
- This is because the "barrier" (in the exponent) for substitutional diffusion has two 'opposing' terms: ΔH_f and ΔH_m (as compared to interstitial diffusion, which has only one term).

For Substitutional Diffusion

$$D = \nu \, \delta^2 \, e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

which is of the form

$$D = D_0 \, e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

- *D (C in FCC Fe at 1000ºC) = 3 × 10^{-11} m^2/s*

For Substitutional Diffusion

$$D = \nu \, \delta^2 \, e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)}$$

which is of the form

$$D = D_0 \, e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)}$$

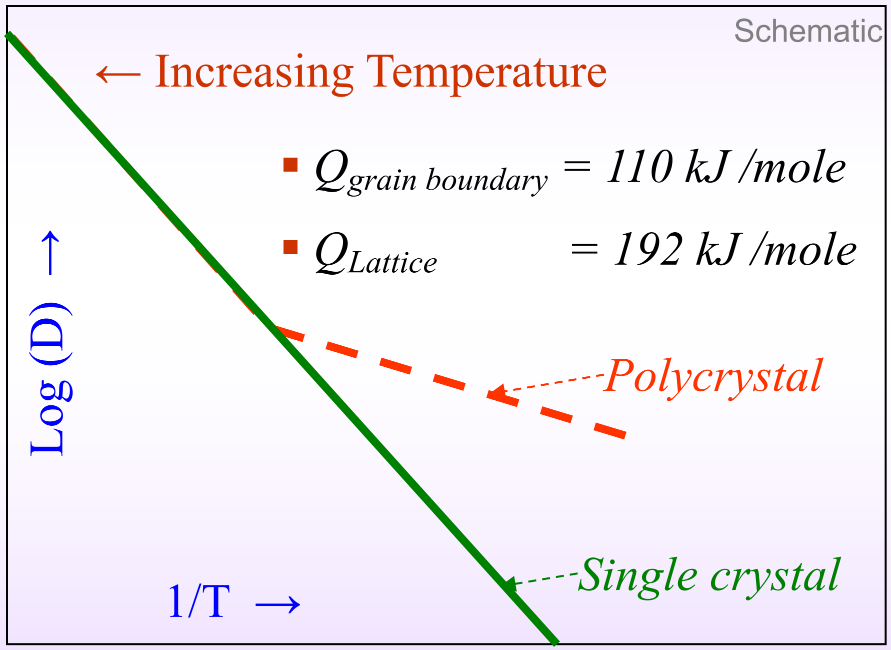
- *D (Ni in FCC Fe at 1000ºC) = 2 × 10^{-16} m^2/s*

Diffusion Paths with Lesser Resistance

- The diffusion considered so far (both substitutional and interstitial) is 'through' the lattice.
- In a microstructure there are many features, which can provide 'easier' paths for diffusion. These paths have a lower activation barrier for atomic jumps.
- The 'features' to be considered include grain boundaries, surfaces, dislocation cores, etc. Residual stress can also play a major role in diffusion.
- The order for activation energies (Q) for various paths is as listed below. A lower activation energy implies a higher diffusivity.
- However, the flux of matter will be determined not only by the diffusivity, but also by the cross-section available for the path.
- The diffusion through the core of a dislocation (especially so for edge dislocations) is termed as Pipe Diffusion.

Experimentally determined activation energies for diffusion $Q_{surface} < Q_{grain\ boundary} < Q_{pipe} < Q_{lattice}$

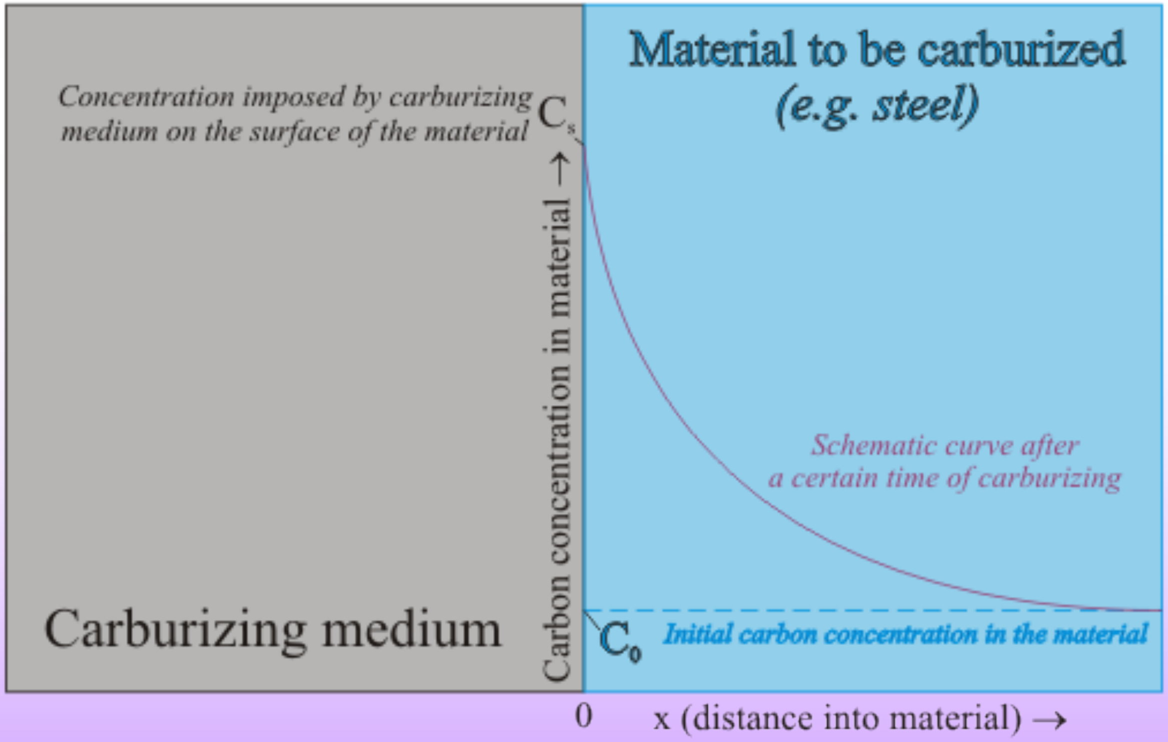
❑ If the 'true' effect of the high diffusivity of a low cross-section path is to be observed, then we need to go to low temperatures. At low temperatures, the high activation energy (low diffusivity) path is practically frozen and the effect of low activation energy path can be observed.



Comparison of Diffusivity for self-diffusion of Ag → single crystal vs polycrystal

Applications based on Fick's II law → *Carburization of steel*

- Surface is often the most important part of the component, which is prone to degradation.
- Surface hardening of steel components like gears is done by carburizing or nitriding.
- Carburizing is done in the γ-phase field, where in the solubility of carbon is higher that that in the α phase. The high temperature enhances the kinetics as well.
- In pack carburizing, a solid carbon powder used as C source.
- In gas carburizing Methane gas is used a carbon source using the following reaction.
  CH_4 (g) → $2H_2$ (g) + C (the carbon released diffuses into steel).



It is usually assumed that the carbon concentration on the surface (C_S) is constant (i.e. the carburizing medium imposes a constant concentration on the surface). An uniform homogeneous carbon concentration (C_0) is assumed in the material before the carburization. Transient diffusion conditions exist and C diffuses into the steel.

We already have the error function solution to the diffusion equation.

$$C(x,t) = A - B \; erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

Using the B.C. we can get the specific solution for the current case (i.e. the values of A & B).

- $C(+x, 0) = C_0$
- $C(0, t) = C_S$

$\downarrow$

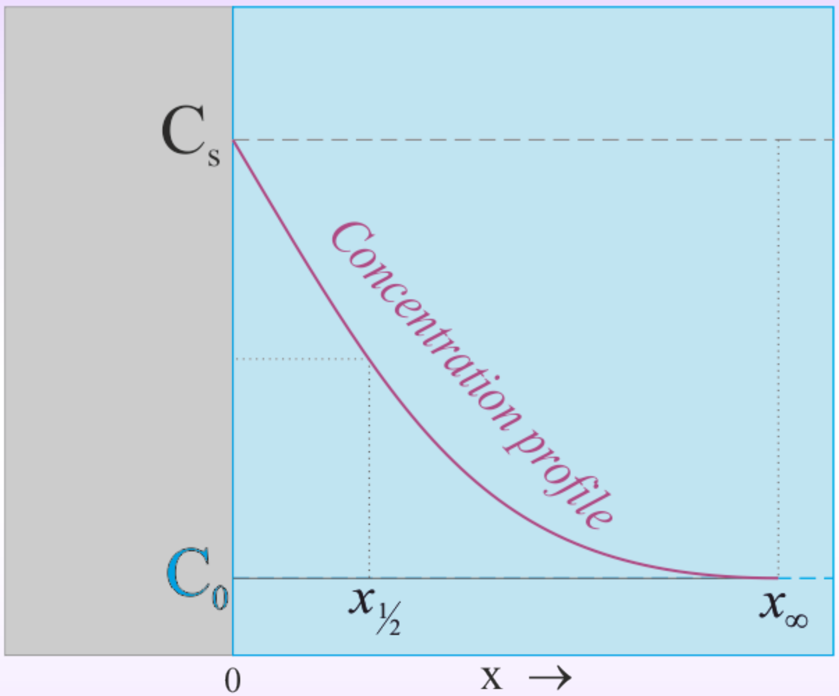
- $A = C_S$
- $B = C_S - C_0$

Hence, the solution is as below.

$$\boxed{\frac{C(x,t) - C_s}{(C_0 - C_s)} = erf\left(\frac{x}{2\sqrt{Dt}}\right)}$$

Approximate formula for depth of penetration

- Often we would like to work with approximate formulae, which tell us the 'effective' depth of penetration and the depth which remains un-penetrated.



$$\left(\frac{C(x,t)-C_S}{C_0-C_S}\right)=erf\left(\frac{x}{2\sqrt{Dt}}\right) \qquad \left(\frac{c(x,t)-C_0}{C_S-C_0}\right)=1-erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

Let the distance at which $[(C(x,t)-C_0)/(C_S-C_0)] = ½$ be called $x_{1/2}$ (which is an 'effective penetration depth')

$$\frac{1}{2}=1-erf\left(\frac{x_{1/2}}{2\sqrt{Dt}}\right) \qquad erf\left(\frac{x_{1/2}}{2\sqrt{Dt}}\right)=\frac{1}{2}$$

$$erf\left(\frac{1}{2}\right)\approx\frac{1}{2} \quad\Rightarrow\quad \left(\frac{x_{1/2}}{2\sqrt{Dt}}\right)=\frac{1}{2}$$

$$\boxed{x_{penetration}=\sqrt{Dt}} \qquad \boxed{x_{1/2}=\sqrt{Dt}}$$

The depth at which $C(x)$ is nearly C_0 is (i.e. the distance beyond which is '*un'-penetrated*):

$$0=1-erf\left(\frac{x_\infty}{2\sqrt{Dt}}\right) \qquad \text{Erf(u)} \sim 1 \text{ when u} \sim 2 \qquad \left(\frac{x_\infty}{2\sqrt{Dt}}\right)=2 \qquad x_\infty=4\sqrt{Dt} \;\Rightarrow\; x_\infty=4x_{1/2}$$

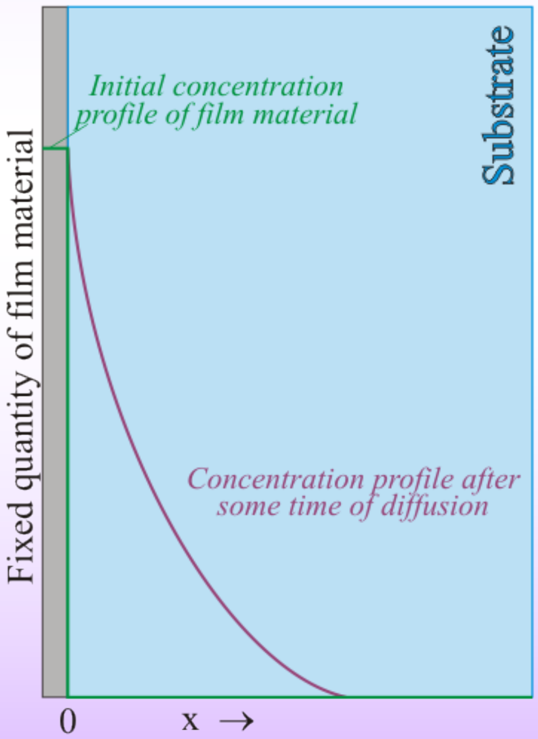
Funda Check

What is the difference between fluid flow and diffusion? Do both of them not involve 'mass flow'.

- Let us look at 'schematic' illustrative pictures as below. In diffusion, the motion of specific species of matter (say atoms, molecules, ions,...) with respect to a surrounding background (which is also ofcourse matter!).

Another solution to the Fick's II law

- A thin film of material (fixed quantity of mass M) is deposited on the surface of another material (e.g. dopant on the surface of a semi-conductor). The system is heated to allow diffusion of the film material into the substrate.
- For these boundary conditions we can use an exponential solution.



Boundary and Initial conditions

Initially the species is only on the surface → $C(+x, 0) = 0$

The total mass of the species remains constatant → $$\int_0^\infty c\,dx = M$$

The exponential solution → $$c(x,t) = \frac{M}{\sqrt{\pi D t}} \exp\left(\frac{-x^2}{4Dt}\right)$$

Diffusion in ionic materials

- Ionic materials are not close packed (like CCP or HCP metals).
- Ionic crystals may contain connected void pathways for rapid diffusion.
- These pathways could include ions in a sublattice (which could get disordered) and hence the transport is very selective
  - β alumina compounds show cationic conduction
  - Fluorite like oxides are anionic conductors.
- Due to high diffusivity of ions in these materials they are called superionic conductors. They are characterized by:
  - High value of D along with small temperature dependence of D
  - Small values of D_0.
- Order disorder transition in conducting sublattice has been cited as one of the mechanisms for this behaviour.

Calculated and experimental activation energies for vacancy Diffusion

Element	ΔH_f	ΔH_m	$\Delta H_f + \Delta H_m$	Q
Au	97	80	177	174
Ag	95	79	174	184

End

Solved Example

A 0.2% carbon steel needs to be surface carburized such that the concentration of carbon at 0.2mm depth is 1%. The carburizing medium imposes a surface concentration of carbon of 1.4% and the process is carried out at 900°C (where, Fe is in FCC form).

Data: $D_0(\text{C in } \gamma\text{-Fe}) = 0.7 \times 10^{-4}\ \text{m}^2/s$ $Q = 157\ kJ/mole$

Given: T = 900° C, $C_0 = C(x, 0) = C(\infty, t) = 0.2$ % C,

$C_f = C(0.2\text{ mm}, t_1) = 1\%$ C (at x = 0.2 mm), $C_s = C(0, t) = 1.4\%$ C

The solution to the Fick' second law: $$C(x,t) = A - B\ erf\left(\frac{x}{2\sqrt{Dt}}\right) \quad (1)$$

The constants A & B are determined from boundary and initial conditions:

$$C(0,t) = A = C_S = 0.014\,,\quad C(\infty,t) = A - B = C_0 = 0.002 \text{ or } C(x,0) = A - B = C_0 = 0.002$$

$$\text{B} = \text{C}_\text{S} - \text{C}_0 = 0.012\,,\ C(x,t) = 0.014 - 0.012\ erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\boxed{C(x,t) = \text{C}_\text{S} - (\text{C}_\text{S} - \text{C}_0)\ erf\left(\frac{x}{2\sqrt{Dt}}\right)}$$

$$\boxed{\left(\frac{C(x,t) - C_S}{C_0 - C_S}\right) = erf\left(\frac{x}{2\sqrt{Dt}}\right)}$$

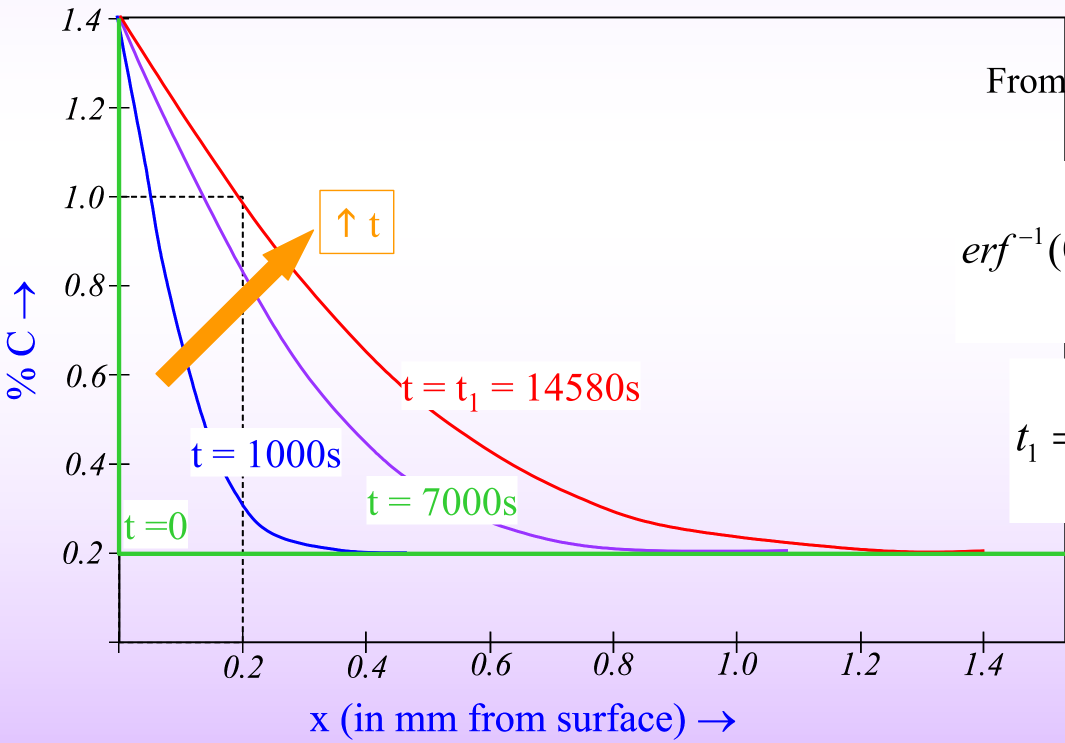
$$C(2 \times 10^{-4} m, t_1) = 0.01 = 0.014 - 0.012\ erf\left(\frac{2 \times 10^{-4}}{2\sqrt{Dt_1}}\right)$$

$$\frac{1}{3} = erf\left(\frac{2 \times 10^{-4}}{2\sqrt{Dt_1}}\right) \quad (2)$$

The following points are to be noted:

- The mechanism of C diffusion is interstitial diffusion
- The diffusivity 'D' has to be evaluated at 900°C using: $D = D_0 \exp\left(\frac{-Q}{RT}\right)$

$$D = D_0 \exp\left(\frac{-Q}{RT}\right) = (0.7\times10^{-4})\exp\left(\frac{-157\times10^{3}}{8.314\times1173}\right) = 7.14\times10^{-12}\ \frac{m^2}{s}$$

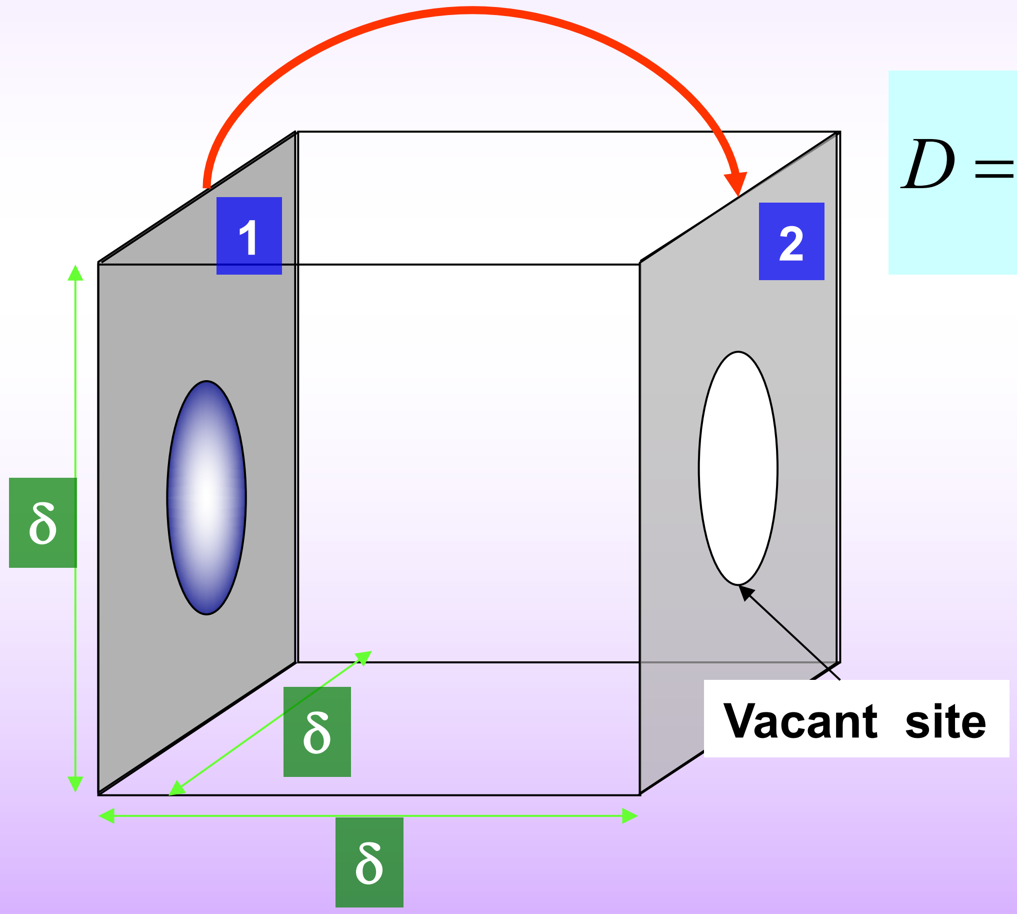


From equation (2) $\frac{1}{3} = erf\left(\frac{2\times10^{-4}}{2\sqrt{Dt_1}}\right)$

$$erf^{-1}(0.3333) \approx 0.309 = \left(\frac{2\times10^{-4}}{2\sqrt{\left(7.14\times10^{-12}\right)t_1}}\right)$$

$$t_1 = \frac{1}{\left(7.14\times10^{-12}\right)}\left(\frac{10^{-4}}{0.33}\right)^2 = 14580\,s$$

- c = atoms / volume
- $c = 1 / \delta^3$
- concentration gradient $dc/dx = (-1 / \delta^3)/\delta = -1 / \delta^4$
- Flux = No of atoms / area / time = ν' / area = ν' / δ^2



$$D = \frac{J}{-(dc/dx)} = \frac{\nu'}{\delta^2}\delta^4 = \nu'\delta^2$$

$$D = \nu\,\delta^2\,e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

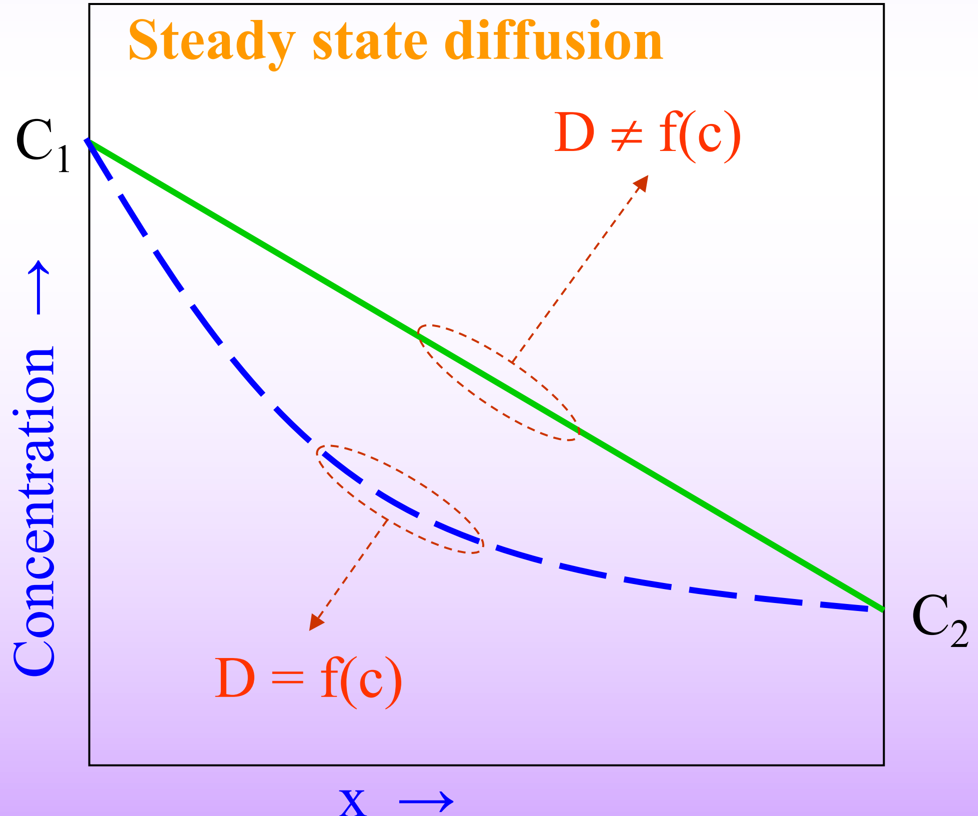
On comparison with

$$D = D_0\,e^{\left(-\frac{Q}{kT}\right)}$$

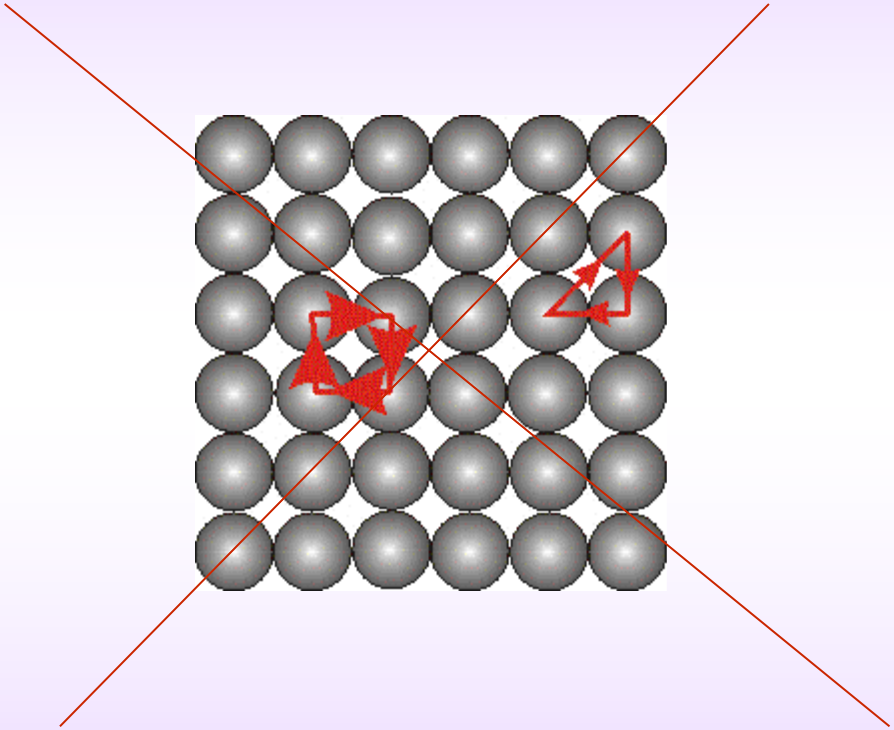
$$D_0 = \nu\,\delta^2$$

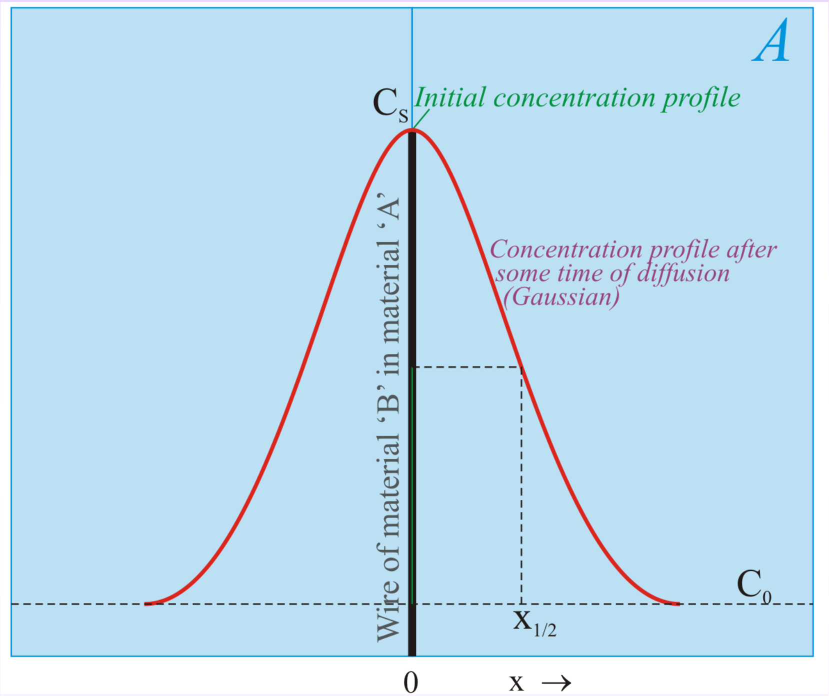
3. Interstitialcy Mechanism

- Exchange of interstitial atom with a regular host atom *(ejected from its regular site and occupies an interstitial site)*
- Requires comparatively low activation energies and can provide a pathway for fast diffusion
- Interstitial halogen centres in alkali halides and silver interstitials in silver halides



4. Direct Interchange and Ring





$$\left(\frac{dc}{dt}\right)\Delta x = -\frac{dJ}{dx}\Delta x$$